

Beomjin Kim

Software Engineer — Backend Systems & Production Reliability

Target role: Senior Application Support Engineer, Sales & Delivery Applications

Total experience 5+ years 010-8849-8539 hypnos709@gmail.com GitHub LinkedIn

Professional Summary

Software engineer with 5+ years of experience, including 3+ years across backend systems, database workflows, production reliability, and operational tooling. I have designed and operated production-facing order, payment, and real-time messaging systems where API contracts, data consistency, observability, and deployment safety had to align. My strongest fit for Tesla is combining backend development with incident-style problem solving, database/log-oriented analysis, and automation that reduces repeated operational work and third-party dependency cost.

Fit for Tesla Application Support													
Production Issue Analysis Redesigned a payment flow after analyzing double-payment, missing-order, timeout, and data consistency risks.	Sales / Order Workflow Experience Built order creation, payment verification, refund policy, transaction handling, and status tracking flows.												
SQL / Database-backed Systems Worked with PostgreSQL, Supabase, Oracle redo logs, data synchronization, and data pipelines.	Support Tooling and Automation Built development automation, test generation flows, documentation automation, and internal viewer/tooling around logs and workflows.												
Real-time System Design Deployed and operated an in-house chat service in production with load-tested scalability, CloudWatch observability, blue-green deployment, and no server outage over 2–3 months.													
Skills <table><tr><td>Application Support</td><td>Production issue analysis, root-cause-oriented debugging, containment design, long-term resolution planning, cross-functional communication</td></tr><tr><td>Backend / Data</td><td>NestJS, FastAPI, PostgreSQL, Supabase, REST API design, data modeling, transaction and payment state design, real-time messaging architecture, SQL querying and analysis</td></tr><tr><td>Frontend / Mobile</td><td>Flutter, React, Next.js, TypeScript, Riverpod, BLoC, React Query, responsive web, shared UI systems</td></tr><tr><td>Platform / Infra</td><td>AWS, Firebase, GitHub Actions, Fastlane, Socket.IO, Redis Pub/Sub, Kafka, ECS Fargate, networking fundamentals, databases, CI/CD, application monitoring concepts</td></tr><tr><td>Automation</td><td>Python, C, Claude API, MCP, AI Agent Harness, Swagger-based code generation, test automation, documentation automation</td></tr><tr><td>Languages</td><td>Korean: Native English: Working proficiency (computer science coursework and technical documentation)</td></tr></table>		Application Support	Production issue analysis, root-cause-oriented debugging, containment design, long-term resolution planning, cross-functional communication	Backend / Data	NestJS, FastAPI, PostgreSQL, Supabase, REST API design, data modeling, transaction and payment state design, real-time messaging architecture, SQL querying and analysis	Frontend / Mobile	Flutter, React, Next.js, TypeScript, Riverpod, BLoC, React Query, responsive web, shared UI systems	Platform / Infra	AWS, Firebase, GitHub Actions, Fastlane, Socket.IO, Redis Pub/Sub, Kafka, ECS Fargate, networking fundamentals, databases, CI/CD, application monitoring concepts	Automation	Python, C, Claude API, MCP, AI Agent Harness, Swagger-based code generation, test automation, documentation automation	Languages	Korean: Native English: Working proficiency (computer science coursework and technical documentation)
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Professional Experience

2024.12 - Present

Munto

Full-stack Engineer

Worked across web, mobile, backend order/payment flows, common modules, and development automation while the product expanded from mobile-first usage to web and multi-service experiences.

Web MAU 80K to 120K, total MAU 250K+

Production Reliability / Payment Flow

- Analyzed double-payment and payment-completed/order-missing risks caused by client-side payment confirmation and unstable order state transitions.
- Redesigned the flow so the server owns order creation, payment verification, transaction handling, refund policy, and final payment state decisions.
- Introduced a WAITING_PAYMENT state and order-level idempotency so retry and re-entry flows resolve against the same order instead of creating inconsistent payment attempts.
- Handled payment provider confirm timeouts with PROCESSING responses and order-status polling instead of prematurely failing user-visible payment flows.
- After rollout, no duplicate-payment incident attributable to the redesigned internal order/payment flow recurred; payment-provider incidents were isolated and handled separately.
- Supported payment operations for a service with approximately 15K total DAU and 8,000 weekly applications, currently processing KRW 600–700M in monthly transaction volume and reaching a peak of KRW 1B immediately after rollout.

Application Architecture / Business Operations

- Separated server state with React Query cache and local UI state with hooks and Context API to reduce coupling between business data and screen state.
- Built and deployed an in-house chat service to production, replacing Sendbird, with separate REST/WebSocket deployments, PostgreSQL persistence, Redis Pub/Sub multi-node synchronization, message ordering, read-cursor unread counts, and push SLA design.
- Load-tested the architecture for workloads targeting up to 100,000 users and configured AWS CloudWatch observability and blue-green deployments; operated it for 2–3 months without a server outage.
- Migrated the first segment of approximately 500 active chat rooms, reducing SaaS cost by approximately \$300/month and creating a path to an estimated additional \$1,700/month reduction by migrating 30–50% of total chat usage.
- Designed the munto-core common package for shared network, authentication, token, storage, logging, internationalization, Firebase, and chat SDK initialization modules.

Automation / Process Improvement

- Established a standard development process from planning/SRS to WBS, DBML, Swagger, TCL, implementation, and verification.
- Built an AI agent harness that reproduces the same development workflow across Claude, Cursor, and Codex, improving deliverable completion time by approximately 60%.

2022.09 - 2024.03

Cenacle Soft

Flutter Engineer

Built healthcare data and EMR-integrated application workflows from initial launch.

- Developed healthcare data, EMR integration, hospital reservation, and mobile patient chart workflows.
- Separated server DTOs from application models and designed factory-based domain models for heterogeneous health data.

2020.04 - 2022.04

WeDataLab

Software Engineer

Worked on DB monitoring solutions, system-level communication modules, internal tools, and Flutter applications.

- Developed an Oracle redo log-based data synchronization module for database monitoring and replication-related workflows.
- Managed Oracle Pro*C build and executable generation flows.
- Built socket communication and semaphore-based log transfer modules in C.
- Developed a Python/PyQt log viewer tool for internal inspection and debugging workflows.

Education

2016.03 - 2023.08

Sungkyunkwan University

B.A. in Philosophy, B.S. in Computer Science and Engineering

Keywords: application support, production support, order/payment systems, SQL, database analysis, REST APIs, real-time messaging, Redis, Socket.IO, incident analysis, internal tools, automation, cross-functional support.